LCS 100
Installation Guide

webfleet solutions

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Installation

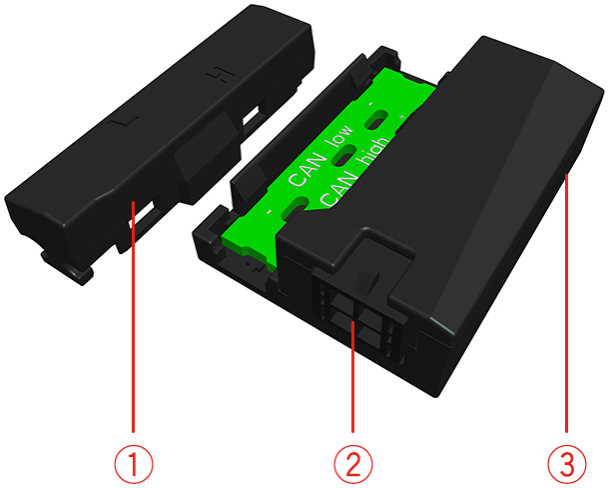
Read me first

Congratulations

Congratulations! You have chosen the LCS 100, a key accessory for the LINK 610/710/710 B. The LCS 100 is a CAN sensor that lets your LINK 610/710/710 B read CAN bus information using a contactless connection between your LINK 610/710/710 B and the CAN bus of your vehicle.

Using CAN bus data you can make use of more features that may be supported by your Webfleet Solutions products, such as fuel consumption reporting or vehicle maintenance monitoring.

What’s in the box



- LCS 100

  1. LCS 100 CAN Cable Clip with markings for CAN high (H) and CAN low (L)
  2. 4-pin Data/Power Cable connector
  3. LED indicating data traffic.

- Data / Power cable

  - Basic installation guide

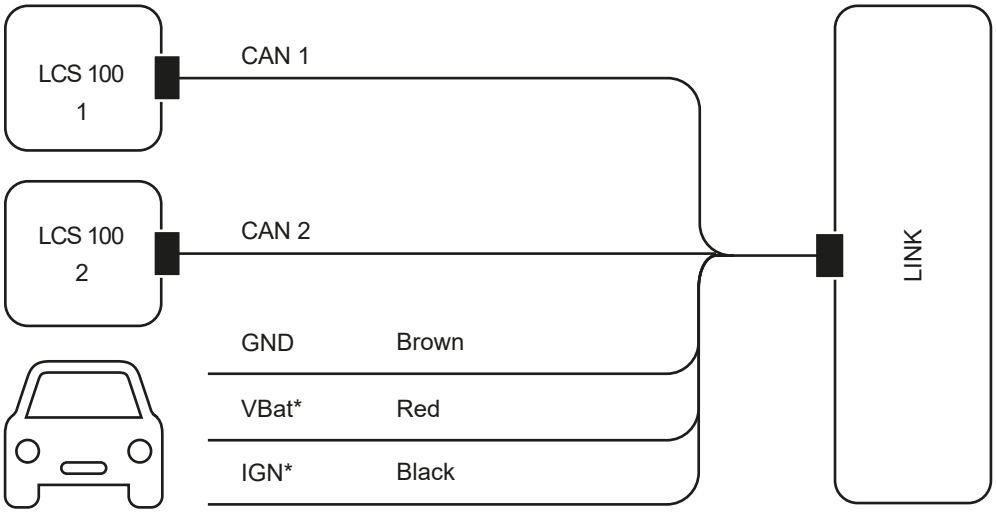
What you need for the installation

Before starting the installation of your LCS 100, read the installation instructions and the safety notices and warnings carefully and make sure you have the following:

- All parts contained in the box.
- A LINK 610/710/710 B either to be installed or already installed in the vehicle.
- A LINK 610/710/710 B installation guide for testing operation.
- Access to the Webfleet Solutions installation tools and access to the LCS 100 installation location in the vehicle that is described there.

Connection overview

Connection overview



* Make sure this wire is fused with max. 10A.

Note: Only LINK devices listed in Accessories not supplied with this device are supported.

Installing LCS 100

To install your LCS 100 do the following:

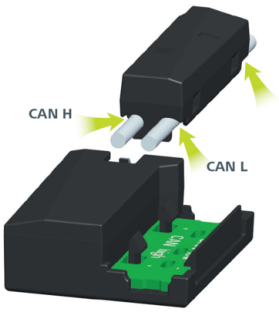
1. Find the correct installation location in your vehicle for the CAN bus that you want to read from (CAN 1 and/or CAN 2).

   Note: For the correct installation location in your specific vehicle, refer to the Webfleet Solutions installation tools or contact the Webfleet Solutions Customer Support for assistance.

2. Identify the CAN high wire.
3. Identify the side of the CAN Cable Clip that is reserved for the CAN high wire.

   Note: The marking **H** on the top side of the CAN Cable Clip indicates the side that is reserved for the CAN high wire.

4. Place the CAN high wire lengthwise on the side that is reserved for the CAN high wire.
5. Push the wire on both ends of the CAN Cable Clip under the cable holder noses.
6. Repeat the previous steps for the CAN low wire.
7. Pull the cable tightly so that the wires in the clip are straight.

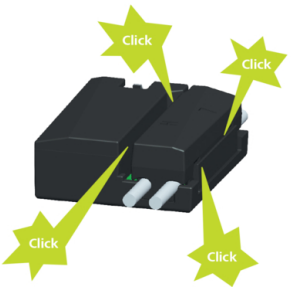


8. Place the clip on the sensor so that the cables for CAN low and CAN high are located on the corresponding sensor sides.

   Note: Refer to the descriptions **CAN high** and **CAN low** on the sensors.

9. Press the clip onto the sensor until all four retaining hooks click into place.

   IMPORTANT: Make sure all four retaining hooks are clicked into place.



10. Depending on which CAN bus wires you have inserted in the CAN Cable Clip, select the 4-pin plug from the Data/Power Cable for **CAN 1** or **CAN 2** and insert it in the connector of the LCS 100.
11. If your vehicle requires more than one CAN bus connection, repeat the steps described before for the second CAN bus connection.

12. If the LINK 610/710/710 B that you want to connect the LCS 100 to is already fully installed in your vehicle, remove its power connection cable completely from the vehicle.
13. Connect the LINK 610/710/710 B to power using the brown, black and red wire of the LCS 100 Data/Power Cable as described in the LINK 610/710/710 B installation guide in the section **Connecting to power**.
14. Verify the installation by following the steps in the section **Testing operation** of the LINK 610/710/710 B installation guide.

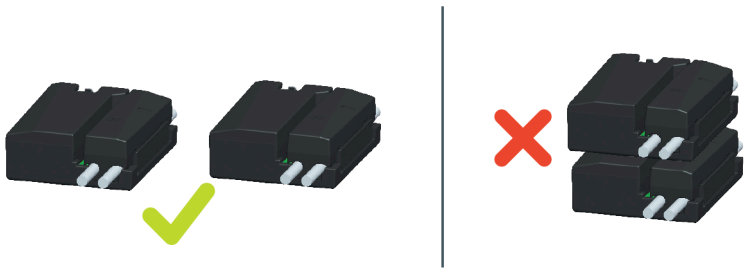
Tips for the installation

Removing the CAN Cable Clip

If you need to remove the CAN Cable Clip from the LCS 100 insert a thin screwdriver from above into the hole for one of the retainer hooks between the sensor and the clip and slightly push it until the clip loosens.

Avoiding interferences and malfunctions

To avoid interferences and malfunctions make sure to install the LCS 100 with a distance of at least 15 cm/6 inch from another CAN bus and/or another LCS 100.



To avoid interference with the 4-wire CAN sensor cable, consider the following: Do not run the 4-wire cable parallel to automotive wiring harnesses. Lead it directly to the LINK device. If necessary, bundle the remaining cable and fix it at a **minimum distance of 5 cm/ 2 inch** from car wiring harnesses or other electrical devices.

Technical data

Dimension	40.6 mm x 30 mm x 12.6 mm (L/W/H) 1.6 inch x 1.18 inch x 0.5 inch (L/W/H)
Weight	12 g / 0.42 ounces
Material	PC/ABS
Protection class	IP20
Supply voltage	5 V (min. 4 V to max. 8 V)
Current/power consumption	At 5 V: typically <= 0.0027 A / 0.0135 W Standby: typically <= 0.00055 A / 0.00275 W
Output voltage	Low: 0 V (-0.1 V) High: 3.8 V (+/- 0.15V), CMOS logic level
Temperature	Operation: -30 °C to +70 °C Storage: -40 °C to +85 °C

Addendum

Important safety notices and warnings

Webfleet Solutions accept no liability for damage that results from disregarding the safety instructions. This document is part of the product. Keep it in a safe place. If you pass the unit on to a new user, make sure you give them this document as well.

Positioning

Position the device in such a way that it doesn't block or interfere with your view of the road, the controls in your vehicle, or the rear-view mirrors and airbags.

Danger of explosion

Parts of this product can cause sparks that can lead to explosions. This may endanger human health and life. Do not use the unit in areas with high risk of explosion. When using this product in a vehicle fuelled by liquefied gas, follow the safety regulations of the country in which the vehicle is operated.

Damage caused through improper installation

The installation and initial operation of the unit must be performed by authorised personnel only, for example, a qualified radio dealer or an automotive electronics workshop.

Risk of injury in case of accidents

Do not mount the unit or accessories in the inflation area of airbags or in the impact area for the head or knees. Choose an installation location that will avoid interference with displays, safety equipment and controls.

Damage to the chassis

Make sure you do not drill into parts of the chassis that have structural or security-related functions. This is because you cannot be certain that they will function properly after modification.

Risk of fire

Make sure you do not drill into covered wiring harnesses, fuel lines or similar components. Drilling into these can cause fire.

Repair and replacement

Repairs must be carried out by authorised and qualified personnel only. Never replace damaged parts of the unit yourself. Send the defective unit to Webfleet Solutions for repair. Only the qualified staff of Webfleet Solutions are authorised to repair or replace parts.

CE marking

CE

This device can be used in all EU Member States.

Hereby, Webfleet Solutions declares that the electronic equipment accessory LINK CAN Sensor 100 is in compliance with Directive 2014/30/EU (EMC). The full text of the EU declaration of conformity is available at the following internet address: https://www.webfleet.com/webfleet/legal/doc/.

Usable with

LINK 610/710/710 B (and requires a LINK CAN Sensor 100 Data / Power cable)

Operating temperature

This device will remain fully operational within the following temperatures: -30 °C to +70 °C

Prolonged exposure to higher or lower temperatures can cause damage to your device and is therefore advised against.

WEEE – e-waste disposal

As required by Directive 2012/19/EU (WEEE) this product is marked with the crossed-out wheeled-bin symbol on its body and/or packaging. This product shall not be treated as household waste or be disposed of as unsorted municipal waste. To help protect the environment, please dispose of this product by returning it to the point of sale or bringing it to your local municipal collection point for recycling

Triman logo

This document

Great care was taken in preparing this document. Constant product development may mean that some information is not entirely up to date. The information is subject to change without notice. Webfleet Solutions shall not be liable for technical or editorial errors or omissions contained herein, nor for incidental or consequential damages resulting from the performance or use of this document. This document may not be copied without the prior written consent from Webfleet Solutions B.V.

Technical Specifications

Supply voltage

5 V (min. 4 V to max. 8 V)

Maximum power consumption

At 5 V: 0.0135 W

Accessories supplied with this device

- Data/Power cable
- CAN Cable Clip
- Basic installation guide

Accessories not supplied with this device

- LINK 610/710/710 B

Terms and conditions: Limited warranty and EULA reference

Our terms and conditions, including our limited warranty and end user license terms apply to this product; visit https://www.webfleet.com/legal

Copyright notices

© 2021 Webfleet Solutions. All rights reserved. WEBFLEET is a registered trademark of Webfleet Solutions B.V. or one of its Affiliates.

Webfleet Solutions Limited Warranty

WARRANTOR

Non-U.S. and non-Canadian purchases: If you have made your purchase outside the United States and Canada, this Limited Warranty is granted by and this Limitation of Liability is stipulated for the benefit of Webfleet Solutions B.V., De Ruijterkade 154, 1011 AC Amsterdam, The Netherlands.

WHAT THIS WARRANTY COVERS

1 Webfleet Solutions B.V. ("Webfleet Solutions") warrants to you that the Hardware will be free from defects in workmanship and materials under normal use ("Defects") for a period of one (1) year from the date that the Hardware was first purchased by you ("Warranty Period"). During the Warranty Period the Hardware will be repaired or replaced at Webfleet Solutions' choice ("Limited Warranty") without charge to you for either parts or labour. This Limited Warranty covers the replacement of the Hardware only. If the Hardware is repaired after the Warranty Period has expired, the Warranty Period for the repair will expire six (6) months after the date of repair.

WHAT THIS WARRANTY DOES NOT COVER

2 The Limited Warranty does not apply to normal wear and tear, does not apply when the Hardware is opened or repaired by someone not authorized by Webfleet Solutions and does not cover repair or replacement of any Hardware or part thereof damaged by: misuse, moisture, liquids, proximity or exposure to heat and accident, abuse, non-compliance with the instructions supplied with the Hardware, neglect or misapplication. The Limited Warranty does not cover physical damage to the surface of the Hardware. This Limited Warranty does not cover any software that may accompany or be installed on the Hardware. The Limited Warranty does not cover the installation, removal or maintenance of the Hardware or any costs related herewith.

HOW TO MAKE A WARRANTY CLAIM

3 In order to make a claim of a Defect, you must contact Webfleet Solutions during the Warranty Period via www.webfleet.com to explain the Defect and to obtain an RMA number (Return Materials Authorization) if necessary. You must return the Hardware during the Warranty Period, along with the RMA number provided by Webfleet Solutions and an explanation of the Defect, to the address provided to you by Webfleet Solutions. If a defect arises and a valid claim under this Limited Warranty is received by Webfleet Solutions after the first one hundred and eighty (180) days of the Warranty Period, Webfleet Solutions is entitled to charge you for any reasonable shipping and handling costs made in connection with the repair or replacement of the Hardware. You must comply with any other return procedures stipulated by Webfleet Solutions, if any.

YOUR LEGAL RIGHTS

4 Some countries may not allow the exclusion or limitation of damages. If any part of this Limited Warranty is held to be invalid or unenforceable, the remainder of the Limited Warranty shall nonetheless remain in full force and effect.

5 This Limited Warranty is the only express warranty made to you and is provided in lieu of any other express warranties or similar obligations (if any) created by any advertising, documentation, packaging, or other communications.

6 Except for the Limited Warranty and to the maximum extent permitted by applicable law, Webfleet Solutions and its suppliers provide the Hardware "AS IS AND WITH ALL FAULTS", and hereby disclaim all other warranties and conditions, whether express, implied or statutory, including, but not limited to, any (if any) implied warranties, duties or conditions of satisfactory quality, of fitness for a particular purpose, of reliability or availability, of accuracy or completeness of responses, of results, of workmanlike effort, of lack of viruses, and of reasonable care and skill, all with regard to the Hardware, and the provision of or failure to provide support or other services, information, software, and related content through the Hardware or otherwise arising out of the use of the Hardware. Also, there is no warranty or condition of quiet enjoyment, quiet possession, or non-infringement with regard to the Hardware. This exclusion does not apply to

(I) any implied condition as to title and

(II) any implied warranty as to conformity with description.

7 This Limited Warranty does not affect any legal rights under applicable national legislation governing the sale of consumer goods.

8 This Limited Warranty cannot be transferred to any other person.

LIMITATION OF LIABILITY

9 Neither Webfleet Solutions nor its suppliers shall be liable to you or to any third party for any damages either direct, indirect, incidental, consequential or otherwise (including in each case, but not limited to, damages for the inability to use the equipment or access data, loss of data, loss of business, loss of profits, business interruption or the like) arising out of the use of or inability to use the Hardware even if Webfleet Solutions has been advised of the possibility of such damages.

10 Notwithstanding any damages that you might incur for any reason whatsoever (including, without limitation, all damages referenced herein and all direct or general damages in contract or anything else), the entire liability of Webfleet Solutions and any of its suppliers shall be limited to the amount actually paid by you for the Hardware.

11 Webfleet Solutions shall not be liable for

(I) any fraud on the part of its employees and/or agents; or

(II) any fraudulent misrepresentation on the part of its employees and/or agents.

12 Notwithstanding the above, neither party's liability for death or personal injury resulting from its own negligence shall be limited.